Query Analytics Workloads

4-Day Team Sprint Plan

One person per day — finish your phase, then relax.
Each handoff unlocks the next team member.

DAY 1	DAY 2	DAY 3	DAY 4
Person 1	**Person 2**	**Person 3**	**Person 4**
Data Engineering	**EDA & Visualization**	**Model Training**	**Validation & Report**

Data Cleaning → **EDA & Viz** → **Modeling** → **Validation + Report**

Dataset: UCI ML Repository #493 — Query Analytics Workloads | R (Primary) + Python (Secondary) | April 2026

Day 1

Person 1

Data Engineer & Preprocessor

Lay the foundation — everyone builds on your work.

Morning — 9:00 AM

+ Download all 3 UCI #493 sub-datasets: Range Queries Aggregates, Radius Queries, Radius Queries Count
+ Set up shared repo: folders data/raw, data/processed, notebooks/, report/, slides/
+ Write requirements.txt (Python) and renv.lock (R) with all project dependencies
+ Parse all CSVs in R (readr) and Python (pandas) — verify row counts match documentation

Midday — 12:00 PM

+ Schema validation: verify column types, value ranges, and confirm no null values exist
+ Compute full descriptive statistics: min, max, mean, median, SD, skew, kurtosis for all columns
+ Identify and remove exact duplicate rows — document retention rate in a processing log
+ Check for boundary effects: queries near the Chicago spatial edge with truncated coverage areas

Afternoon — 3:00 PM

+ Engineer base spatial features: query area (pi x R^2 for discs; 4 x Xrange x Yrange for rectangles)
+ Compute normalized X/Y coordinates to [0, 1] relative to full dataset spatial bounds
+ Compute distance of each query center from Chicago's geographic centroid
+ Apply log-transform to Count and SUM targets — export both raw and log-transformed versions

End of Day — 5:30 PM

+ Export 3 clean CSVs to data/processed/: clean_range_queries, clean_radius_queries, clean_radius_count
+ Write 01_preprocessing.Rmd with inline commentary and pipeline diagram placeholder
+ Update README.md with status note (3 sentences) and any data anomalies found
+ Draft Data Processing section (~250 words) for report Section 3 — pipeline steps and assumptions

DELIVERABLES	HANDOFF NOTE
+ data/processed/ — 3 clean CSVs with schema summary + 01_preprocessing.Rmd + Python equivalent notebook + README.md updated with status note and anomalies + Data Processing section draft (report Section 3)	*Commit clean CSVs and preprocessing notebook by 5:30 PM. Update README with a 3-sentence status note. Person 2 loads your outputs first thing on Day 2.*

Relax tonight. Your clean data unblocks the whole team.

Day 2

Person 2

EDA & Visualization Specialist

Tell the story hiding in the data — your charts go in the final report.

Morning — 9:00 AM

- Load Person 1's clean CSVs — run a quick schema check to confirm handoff integrity
- Distribution analysis: histograms of Count, SUM, AVG for all 3 sub-datasets (raw + log-transformed)
- Side-by-side overlay: compare Count distributions between Range Queries and Radius Queries sub-datasets
- Skewness/kurtosis check — confirm whether log-transform sufficiently normalizes the regression targets

Midday — 12:00 PM

- Spatial heatmap: 2D density plot of query origins (X vs Y) with ggplot2 geom_bin2d() — Figure 1
- Scatter plots: Radius vs Count, Area vs SUM, Area vs AVG — identify non-linear relationships
- Outlier detection: IQR-based and Z-score flagging; mark outlier points on the scatter plots
- Cross-dataset summary table: mean and SD of Count, SUM, AVG for Range vs Radius sub-datasets

Afternoon — 3:00 PM

- Correlation matrix: Pearson and Spearman heatmap of all numeric features using R corrplot — Figure 2
- Identify the top 3 features most correlated with Count — document as key modeling insights
- Boxplots by spatial quadrant: divide Chicago into 4 quadrants, show Count distribution per zone — Figure 3
- Export all figures as PNG (300 DPI) to report/figures/ with consistent names: fig01.png, fig02.png...

End of Day — 5:30 PM

- Write 02_eda.Rmd with narrative commentary on every figure — what it shows and why it matters
- Draft Data Analysis section (~250 words + figure references) for report Section 4
- Update README.md with top EDA findings and key insights from spatial exploration
- Write a brief handoff note for Person 3: list the top 3 features that appear most predictive of Count

DELIVERABLES	HANDOFF NOTE
+ report/figures/ — all EDA figures as PNG at 300 DPI + 02_eda.Rmd — full EDA notebook with figure commentary + Data Analysis section draft (report Section 4) + Written feature summary for Person 3 (top Count predictors)	*Commit all figures and EDA notebook by 5:30 PM. Add a handoff note to README listing the top 3 predictive features. Person 3 starts Day 3 from your findings.*

Relax tonight. Person 3 picks up your figures tomorrow morning.

Day 3

Person 3

Feature Engineer & Model Trainer

Build the models that answer the research question.

Morning — 9:00 AM

+ Load Person 1's clean CSVs — read Person 2's feature priority summary from README
+ Advanced feature engineering: aspect ratio of rectangles; interaction term (area x normalized distance)
+ VIF analysis: detect multicollinearity among spatial features — drop or combine those with VIF > 10
+ 80/20 train/test split with fixed seed (seed=42) — export split indices for full reproducibility

Midday — 12:00 PM

+ Baseline: Linear Regression on raw and engineered features (R: lm(); Python: sklearn LinearRegression)
+ Record train RMSE, test RMSE, MAE, R-squared — save as results/model_linear.csv
+ Intermediate: Random Forest with 500 trees (R: randomForest; Python: sklearn RandomForestRegressor)
+ Extract feature importance rankings from RF — save top features as results/feature_importance.csv

Afternoon — 3:00 PM

+ Advanced: XGBoost regressor targeting Count (primary) and SUM (secondary) regression tasks
+ 5-fold CV grid search over learning_rate, n_estimators, max_depth, subsample — save best params
+ KNN Regressor with K = 5, 10, 20 — select best K by CV RMSE as a spatial proximity baseline
+ Compile all results into results/cv_comparison.csv — RMSE, MAE, R-squared for every model and fold

End of Day — 5:30 PM

+ Export trained model objects: model_linear.pkl, model_rf.pkl, model_xgb.pkl, model_knn.pkl
+ Write 03_modeling.ipynb with full pipeline, CV results table, and feature importance chart
+ Draft Model Training section (~250 words + model comparison table) for report Section 5
+ Update README.md with model training summary, best hyperparameters, and expected performance range

DELIVERABLES	HANDOFF NOTE
+ models/ — 4 trained .pkl model files ready for test-set evaluation + results/ — cv_comparison.csv, feature_importance.csv, per-model metric CSVs + 03_modeling.ipynb — full training notebook with all results + Model Training section draft (report Section 5)	*Commit all model files and results CSVs by 5:30 PM. Person 4 runs final test-set evaluation on Day 4 without any retraining needed.*

Relax tonight. Person 4 validates your models and wraps up the report tomorrow.

Day 4

Person 4

Validation Lead & Report Integrator

Bring it all together — you write the final story and submit.

Morning — 9:00 AM

+ Load all 4 trained model .pkl files — run final predictions on the held-out 20% test set
+ Compute final metrics: RMSE, MAE, R-squared, and MAPE for all 4 models on the test set
+ Build benchmark comparison table: 4 models x 3 metrics showing train and test values side by side
+ Residual plot for XGBoost: fitted vs residuals — check for heteroscedasticity and spatial patterns

Midday — 12:00 PM

+ K-Means clustering: fit K = 3, 4, 5 on (X, Y, normalized area); select K using silhouette score
+ Plot cluster map on 2D spatial grid — colored by cluster label to identify Chicago crime hotspot zones
+ Bias and risk assessment: identify where XGBoost underperforms (large areas, sparse city-edge regions)
+ Write Model Validation (Section 6) and Model Performance (Section 7) — approximately 350 words total

Afternoon — 3:00 PM

+ Integrate all 4 section drafts from Persons 1, 2, and 3 into one cohesive Word document
+ Write Abstract (180 words), Conclusion (180 words), Data Sources, and Bibliography (Chicago/IEEE style)
+ Insert all figures with captions and sequential figure numbers; format all tables consistently
+ Proofread full report for consistent terminology, notation, and citation style throughout

End of Day — 5:30 PM

+ Export final report as PDF — page count check (must be under 5 pages per requirements)
+ Record 10-min video: Person 4 intro+conclusion (3.5 min), P1 pipeline script (2 min), P2 EDA (2 min), P3 models (2.5 min)
+ Tag final repo commit as v1.0 — verify requirements.txt and renv.lock are current and complete
+ Submit: final report PDF, video link, and repository URL to the course submission platform

DELIVERABLES	HANDOFF NOTE
+ Final integrated report PDF (under 5 pages, all 11 sections) + 04_validation.ipynb — clustering and residual analysis notebook + results/final_benchmark.csv — test-set results for all 4 models + 10-minute video presentation — edited and submitted	*Tag v1.0 in the repo. Submit all deliverables. Project complete.*

Submit. The project is complete. Well done, team.